(d) A 0.2% solution of phenol is antiseptic while 1% solution acts as disinfectant.
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

A 0.2% solution of phenol is an antiseptic while 1% solution acts as a disinfectant. Chlorine and Iodine are used as strong disinfectants. Dilute solution of boric acid and hydrogen peroxide are weak antiseptics. Disinfectants harm the living tissues. Hence, statement (b) is not true.

ORGANIC CHEMISTRY

A. Hydrocarbons

1. Which is the fundamental element of all organic compounds?

(a) Nitrogen (b) Oxygen
(c) Carbon (d) Brimstone

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Carbon is the fundamental element of all organic compounds (including bio-compounds). The organic compounds are a large class of chemical compounds in which one or more atoms of carbon are covalently linked to atoms of other elements, most commonly hydrogen, oxygen or nitrogen.

2. A hydrocarbon in which two carbon atoms are joined by a double bond is called as an :

(a) Alkane (b) Alkene
(c) Alkyne (d) Ionic bond
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

A hydrocarbon in which two carbon atoms are joined by a double bond is called as an alkene. Alkenes have the general formula C_nH_{2n}. They are also known as olefins. Ethylene (C_2H_4), Propylene (C_3H_6) and Butylene (C_4H_8) are first three members of this group.

3. Which of the following gases is used in cigarette lighters?

(a) Butane (b) Methane
(c) Propane (d) Radon

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Butane is a flammable hydrocarbon with the molecular formula C_4H_{10}. It is a natural gas perhaps best known for its use as a fuel cigarette lighters. It is also an organic compound known as NGL, a Natural Gas Liquid.

4. Gases used in welding are :

(a) Oxygen and hydrogen (b) Oxygen and nitrogen
(c) Oxygen and acetylene (d) Hydrogen and acetylene
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

Welding is a process of joining two materials, usually metals, by causing coalescence. Acetylene is the most commonly need welding gas it is also used in oxyacetylene welding and cutting. Oxygen is a gas that is used in welding to increase the rate of the welding process.

5. Numbers of sigma and pi bonds in benzene are:

(a) 3, 3 (b) 3, 6
(c) 12, 3 (d) 12, 6
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

Benzene is an important organic chemical compound with chemical formula C_6H_6. Its molecule is composed of 6 Carbon atoms and 6 Hydrogen atoms. Its chemical structure can be described as a hexagonal ring with alternating double bonds. In benzene there are 6 sigma bonds between carbon atoms and 6 sigma bonds between carbon and hydrogen atoms. It has also 3 pi bonds between carbon atoms. Therefore 12 sigma and 3 pi bonds are present in benzene molecule.

B. Alcohol

1. Fermentation of sugar leads to –

(a) Ethyl alcohol (b) Methyl alcohol
(c) Acetic acid (d) Chlorophyll

47th B.P.S.C. (Pre) 2005

Ans. (a)

Ethyl alcohol is formed by the fermentation of sugar, which is made of glucose and fructose.

2. Final product of Fermentation

(a) Pyruvic acid (b) Acetaldehyde
(c) Ethyl alcohol (d) Formic acid

46th B.P.S.C. (Pre) 2004

Ans. (c)

The slow transformation of complex organic matter into simpler substances in the presence of enzymes is called fermentation. In the presence of yeast (which contains the appropriate enzymes) the juices of grapes and other fruits are used for the manufacturing of alcoholic beverages by fermentation. In this process molasses or sugar obtained from sugarcane and fruits or starch obtained from different types of grains, in the presence of enzyme invertase, is first converted into glucose and fructose. The formulas of these two are $C_6H_{12}O_6$.

$$\underset{\text{Sugar}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\text{invertase}} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

Both glucose and fructose are converted into ethyl alcohol (ethanol) and carbon dioxide in the presence of another enzyme, zymase. Both this and zymase enzymes are present in this.

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{zymase}} \underset{\text{Ethyl alcohol}}{2C_2H_5OH} + 2CO_2$$

3. Yeast is used in the production of

(a) sugar (b) hydrochloric acid

(c) alcohol (d) sulfuric acid

(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

Yeast is the most popular baking ingredient. Yeast strains are used to produce alcohol, beer, wine and may be used for pharmaceuticals, flavoring, and even ethanol-based fuels. Fermentation is an essential process for wide range of applications.

4. Glucose is converted to ethyl alcohol by the enzyme –

(a) Maltase

(b) Invertase

(c) Zymase

(d) Diastase

(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

Glucose is converted to ethyl alcohol (ethanol) by the enzyme zymase. Zymase catalyzes the fermentation of sugar (glucose and fructose) into ethanol and carbon dioxide. This enzyme complex naturally occurs in yeast and other anaerobic organisms. This enzyme is used in the preparation of ethanol and alcoholic beverages commercially. Invertase enzyme converts sucrose (cane sugar) into glucose and fructose. Maltase enzyme converts maltose into glucose while diastase enzyme converts starch into maltose.

5. The enzyme which converts glucose into ethyl alcohol is

(a) invertase (b) maltase

(c) zymase (d) More than one of the above

(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (c)

See the explanation of the above question.

6. Nowadays the most popular and safe sugar option is

(a) cyclodextrin (b) aspartame

(c) saccharin (d) fructose

(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

Nowadays the most popular and safe sugar option is aspartame. Aspartame is a low-calorie sweetener that has been used for decades as a way to lower one's intake of added sugars while still providing satisfaction from enjoying something sweet. Aspartame is about many times sweeter than sugar and as such only a small amount as the sweetener is needed to match the sweetness provided by sugar. Aspartame consists of two amino acid–aspartic acid and phenylalanine.

7. The breath test conducted by police to check drunken driver has which one of the following on the filter paper?

(a) Potassium dichromate-sulfuric acid

(b) Potassium permanganate-sulfuric acid

(c) Silica gel coated with silver nitrate

(d) Turmeric

(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

The breath test conducted by police to check drunken driver through old breathalyzer has potassium dichromate-sulfuric acid on the filter paper. When alcohol vapour makes contact with the orange dichromate coated crystals, the colour changes from orange to green due to oxidation of alcohol into acetic acid. The degree of the colour changes is directly related to alcohol level in the breath.

8. Which enzyme converts glucose or fructose to C_2H_5OH?

(a) Invertase

(b) Zymase

(c) Maltase

(d) Diastase

(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)